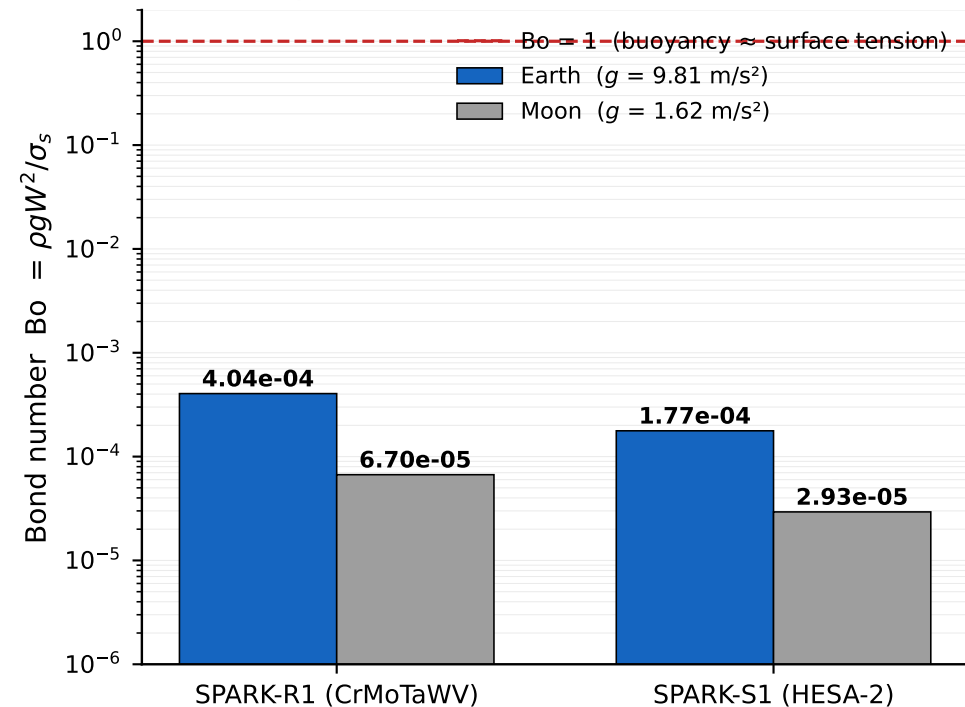


Lunar vs terrestrial LPBF: gravity enters only through the Bond number

Gravity scaling: Earth vs Moon Bond number



Dimensionless regime summary

Quantity	SPARK-R1	SPARK-S1	Implication
Peclet Pe	2.23	5.34	advection / diffusion
Marangoni M	776	1442	thermocapillary scale
Bo (Earth)	$4.04\text{e-}04$	$1.77\text{e-}04$	buoyancy / surface tension
Bo (Moon)	$6.70\text{e-}05$	$2.93\text{e-}05$	reduced by $g_M/g_E \approx 0.17$
Regime	Marangoni	Marangoni	both alloys, both gravities

All four Bond numbers are well below unity, so surface tension dominates over buoyancy in both terrestrial and lunar LPBF for these alloys. Melt-pool free-surface dynamics are Marangoni-driven; lunar operation reduces the (already small) buoyancy contribution by $\sim 6\times$, but does not cross a regime boundary.